

Flooring and Subfloor Build Process

As of: 2026-07-16 **Role:** canonical owner for the truck-bed floor stack, sealing, hardpoint registration, Lonseal installation, and post-install checks.

Owner-confirmed build decision

- The one-piece Lonwood Madera Topseal sheet was glued to the three-piece 3/4 in birch plywood floor with Lonseal #650 on the evening of 2026-07-15 .
- The floor is now intentionally **permanent**. The vinyl bridges both plywood seams; removing the plywood as three sections would require cutting the finish sheet along those seams. Routine floor removal is no longer a planned service path.
- Owner reports mild waviness/bumps and some edge lift. Boards are bolted over the edges for cure pressure. Treat the visible result as a post-cure acceptance/repair decision; do not cut, inject, heat, or aggressively re-roll the floor during cure.
- #650 entered some hardpoint holes. Temporary bolts were driven into the installed stainless rivnuts to preserve the threads. Remove/clean/reinstall those clamp bolts carefully during cure so they do not become permanent or spin/gall an insert.
- The final floor-height limit remains 3/4 in **maximum**. Do **not** add an underlayment layer or another finish-floor layer to mask waviness.
- The stainless rivnuts remain the structural module hardpoints. Furniture still does not anchor to plywood alone.

Current stack

Layer	Material	Working rule
Finish	Lonwood Madera Topseal, Oregano, one piece	Glued 2026-07-15; permanent finish bridging all three plywood sections
Adhesive	Lonseal #650 two-part epoxy	Installed; keep heavy furniture/modules off for 72 hr after actual completion
Subfloor	Three 3/4 in birch plywood sections	Final height-locked substrate; now tied together by the one-piece vinyl
Insulation	5/8 in EPS in bed-rib valleys only	Filler/insulation; trim only where it makes plywood rock or conflicts with a hardpoint
Support	Truck-bed rib highs	The plywood must settle flat on its real bearing points

As-built glue-down and cure gate

- Adhesive completion was reported only as the evening of 2026-07-15 ; the exact clock time is not recorded. The published 72 hr heavy-furniture gate therefore lands no earlier than the corresponding time on 2026-07-18 —use the real completion time if it can be recovered.
- Published #650 handling basis: no foot traffic for 24 hr , no heavy furniture for 72 hr ; the normal reroll window was 2-3 hr after installation and should not be recreated a day later by force.
- While edge boards still need to remain: back out one temporary bolt at a time with a hand ratchet or low-speed driver, stop if a rivnut starts to spin or stainless threads bind, clean adhesive from the bolt/hole, and reinstall only enough to maintain board pressure. Keep any thread lubricant strictly on threads and off the vinyl/plywood bond surface.
- After the heavy-furniture cure window: remove boards, photograph the full floor, inspect every perimeter and hardpoint, identify raised/hollow vs merely wavy areas, and record any board/washer imprinting. Stable cosmetic waviness can be accepted; a loose edge or expanding hollow area needs a localized repair plan before furniture hides it.
- Protect the floor with clean hardboard/cardboard/blankets during every module move. Do not drag extrusion or batteries across the finish.

Practical floor-closeout rules

- Do not increase the floor height.** The current 3/4 in floor is locked.
- Do not structurally fasten furniture only to plywood.** Use the bed rivnuts for the modules that need positive retention. Use plywood-only screws/cleats only where a panel edge or seam needs local flatness control.
- Do not carry bolt clamp load through EPS or an open valley.** Rib-high locations already have solid bed support. Valley locations need a measured rigid path between the plywood underside and the bed/rivnut bearing surface. Rib-edge locations need square bearing or they should be moved/retired.
- Do not bury unknown access needs.** Photograph and dimension installed hardpoint locations before the Lonseal dry fit. Use locator bolts, a template, or two bed datums so finished-floor holes can be recovered without guessing.
- Keep the bed clean and dry before floor closure.** Do not create a formal water-test project; just avoid obvious trapped water, loose debris, lifted tape, and known open dust paths.

Bed sealing approach

The goal is realistic: reduce dust, road splash, and nuisance intrusion under the floor.

- Continue using the Gorilla tape where it is cleanly adhered to the bed liner and is solving an observed seam/gap path.
- Hard-roll the tape, avoid trapped grit, and inspect corners/edges before the plywood goes back down.
- Keep tape, primer, and sealant clear of rivnut threads and the rivnut flange's structural bearing surface.
- Do not intentionally plug an obvious factory drain without a reason. This is a common-sense inspection item, not a requirement to map every body feature.
- After the first dusty or wet drive, inspect accessible floor edges and known former gap areas. Rework only actual failures.

Installed rivnuts and compression-sleeve routing

The rivnuts are now installed. Before floor closure:

1. Run a sacrificial bolt fully in and out of every active insert by hand; flag any spin, damaged thread, poor engagement, or bolt-axis problem.
2. Record each center and classify it as rib high , valley , or rib edge/transition .
3. Confirm the actual module tab/foot can approach the insert without forcing the bolt sideways.
4. Relocate or retire a rib-edge insert if its bolt axis or lower bearing surface cannot be made square. A short sealed filler bolt is acceptable for a retired insert after verifying that it engages without bottoming or protruding into a conflict.

The on-hand stainless sleeves are approximately 0.585 in OD x 7/16 in ID x 5/8 in long . They are **not universal through-plywood compression limiters**: a 5/8 in sleeve is 1/8 in shorter than the 3/4 in plywood before any valley gap is included.

Use them by geometry:

- **Rib high:** plywood already bears on bed metal. Do not place a 5/8 in sleeve under the plywood and do not bore the full plywood thickness for it. Use the real module tab/foot or a broad top washer and tighten only enough to secure the joint without crushing the plywood.
- **Valley/EPS:** this is the useful location for the 5/8 in sleeve, but only if the measured gap from the bed/rivnut bearing surface to the plywood underside is close to the sleeve's actual length. Clear the EPS around the sleeve, keep both sleeve ends square, and verify full lower bearing plus firm contact at the plywood underside. Replace or trim to measured length if the gap differs; do not pull the plywood down to meet a short sleeve or let a long sleeve prop it up.
- **Rib edge/transition:** do not let a square tube land on one edge or a slope. Move/retire the hardpoint or fabricate a properly seated local spacer/saddle only if that location is truly necessary.

If a future hardpoint needs a true sleeve through the plywood, make a custom limiter whose length equals the full distance from the module tab/washer underside to the lower metal bearing surface: 3/4 in plywood + measured under-plywood gap . Bore only after the final hole map is proven. A 5/8 in Forstner hole gives about 0.040 in total clearance around the current 0.585 in sleeve; the top module tab or washer must span that bore positively.

Floor and hardpoint record / remaining sequence

Sections 1-3 preserve the intended pre-install checks for traceability. Glue-down is complete; unchecked historical items are not retroactive holds. Current work starts with the cure/post-cure tasks in Section 4 and restrained reinstall in Section 5.

1. Finish the map

- ☐ Finish only the remaining Desk/storage corrections needed to trust module feet/tabs.
- ☐ Photograph modules, shims, brackets, service openings, and the planned floor-tab locations.
- ☐ Record hardpoint centers from two truck-bed datums.
- ☐ Confirm the known rivnut locations are compatible with underside clearance and module access.

2. Finish sealing and verify installed rivnuts

- ☐ Complete the remaining practical tape/seal pass.
- ☐ Inspect tape adhesion and remove loose debris.
- ☒ Install the stainless rivnuts.
- ☐ Hand-test/map every active insert; classify rib-high, valley, and rib-edge locations.
- ☐ Use the on-hand 5/8 in sleeves only at measured valley/EPS gaps; move/retire angled locations that cannot seat square.
- ☐ Preserve the location map before covering the bed.

3. Settle the existing floor

- ☐ Inspect EPS; trim only proud, wet, damaged, or rocking pieces.
- ☐ Dry-fit the three plywood panels.
- ☐ Eliminate rocking, grit, sharp fasteners, unsupported edges, and visibly moving/high panel joints.
- ☐ Add only the local plywood retention needed to keep an edge or seam quiet and flat.
- ☐ Refit modules with temporary bolts and verify alignment, service access, and battery extraction.

4. Lonseal glue-down — completed / curing

- ☒ Cut and dry-fit the one-piece Lonseal over the assembled three-panel floor.
- ☒ Glue the sheet to the plywood with the complete Lonseal #650 unit on the evening of 2026-07-15 .
- [~] Hold lifted perimeter areas with bolted boards through cure.
- ☐ Back out/clean/reinstall temporary clamp bolts one at a time before adhesive-contaminated threads become permanent; do not use impact force or spin a rivnut.
- ☐ Keep heavy modules/furniture out for 72 hr after the actual adhesive completion time; evening 2026-07-18 is only the earliest planning estimate.
- ☐ Remove cure boards and complete the documented edge/hollow/raised-area/hardpoint inspection before loading the floor.
- ☐ Record accepted cosmetic waviness vs any localized repair area; do not reopen the full floor merely to chase a non-growing cosmetic bump.

5. Reinstall and prove

- ☐ Recover hardpoints from the map; do not blind-drill finished vinyl.
- ☐ Protect Lonseal during module installation.
- ☐ Install module bolts into the registered rivnuts and witness-mark the ones that matter.
- ☐ Confirm no module is loose, rubbing, or blocking electrical/plumbing/battery service access.
- ☐ Reinspect after the first local drive for loose fasteners, floor movement, vinyl indentation, or dust/moisture at known edges.

Simple go / no-go gates

Go when	No-go when
Tape/seal work is adhered and the bed is dry/clean	There is obvious loose tape, trapped debris, standing water, or a known open dust path
Rivnuts set firmly and pass a bolt-in/bolt-out check	An insert spins, has poor thread engagement, or cannot be serviced
Plywood lies flat and its seams do not move under foot	A panel rocks, a seam moves, or EPS holds the plywood high
All module locations are mapped and dry-refit works	Final module bolts or service access would be guesswork after vinyl
Adhesive tools, temperature, mixing, and cure time are ready	Rushing adhesive work would leave the floor unusable or force premature heavy loading

BOM synchronization

BOM owner: bom/bom_estimated_items.csv .

- 141 : 1/4-20 multi-grip rivet-nut tool set with stainless inserts — purchased; truck-bed inserts installed.
- 217 : 5/8 in stainless spacer/compression sleeves — purchased; valley/EPS use only where the measured gap fits.
- 147 : 3/4 in birch base plywood — purchased and height-locked.
- 148 : TotalBoat Halcyon edge/bottom sealer — purchased.
- 149 : Lonwood Madera Topseal sheet — purchased and permanently glued to the three-panel floor on 2026-07-15 .
- 150 : Lonseal #650 0.5 gal — complete unit mixed/used for the 2026-07-15 glue-down; floor curing.
- 174 : exact-notch trowel — purchased.
- 175 : compact roller method — glue-down is complete; exact as-built rolling method/result is not separately recorded.
- 177 : polyurethane sealant — purchased/partially used.
- 178 : Gorilla Waterproof Patch & Seal Tape — bed cavity is now closed by the permanent floor; inspect only if later dust/moisture evidence justifies destructive access.

Do not maintain a second hand-calculated flooring total here; the BOM CSV owns quantity, price, and purchase status.

References

- Lonseal Vehicle Installation Guide (repo copy: `references/lonseal flooring installation guide.pdf`).
- Lonseal #650 Technical Data Sheet: https://lonseal.com/wp-content/uploads/2024/11/TDS_650_110524.pdf.
- Gorilla Waterproof Patch & Seal Tape product page: <https://gorillatough.com/product/gorilla-waterproofing-tape-black/>.

Manufacturer documents guide product handling. This implementation file owns the build-specific decisions: the `3/4 in` height lock, practical tape/seal use, and stainless-rivnut hardpoint plan.